AUDIT TRAIL REVIEW AND MONITORING PROCESS

SOP-014 | Standard Operating Procedure | Issued under POL-014

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Document Control

Field	Detail
Document Title	Audit Trail Review and Monitoring Process
Document ID	SOP-014
Version	1.0 — DRAFT
Parent Policy	POL-014 — Audit Trail and Monitoring Policy
Related Documents	POL-003 (Data Integrity), POL-005 (System Classification), POL-006 (CSV), POL-007 (Change Management), POL-009 (PAM), POL-010 (IS Risk), POL-012 (Incident Response), POL-015 (IT Service Management), SOP-003 (Data Integrity / DIRA), SOP-006 (Validation), SOP-007 (Change Management), SOP-009 (PAM), SOP-010 (IS Risk Assessment), SOP-012 (Incident Response), SOP-015 (ITSM), EXC-001 (Exception Management), RACI-001
Owner	Information Security Manager
Co-Owner	DIO (Data Integrity Officer)
Effective Date	[DATE]
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NOTE: An audit trail that exists but is never reviewed is not a control — it is a record. This SOP is the process that converts audit trail records into operational assurance. The review cadences, alert response timelines, and gap detection procedures in this SOP are what make POL-014's technical requirements meaningful in practice.

Revision History

Version	Date	Author	Description
0.1	[DATE]	[Author]	Initial draft
1.0	[DATE]	[Author]	Issued for review

1. Purpose

This SOP defines the routine processes for reviewing, monitoring, and responding to audit trail records across [Company Name] systems. It operationalises POL-014 by establishing the GxP audit trail review procedure, SIEM alert triage workflow, audit trail gap detection and emergency response, periodic review cadence by system tier, and the immutability verification programme.

This SOP is the primary operational link between POL-014 (Audit Trail and Monitoring Policy) and SOP-003 (Data Integrity Process). Audit trail review is a data integrity quality activity for GxP systems — it is not an IT function. Review records produced under this SOP are GxP quality records, subject to the same version control, retention, and access controls as other GxP documentation.

2. Scope

This SOP applies to audit trail review and monitoring for all systems with a GxP flag, all Tier 1 systems, all SOX ITGC systems, and all Tier 2 systems. Tier 3 and Tier 4 systems are reviewed under the periodic review programme (Section 6) at a reduced cadence.

The SIEM monitoring programme (Section 5) applies to all systems that forward logs to the SIEM — which includes all Tier 1/2 and all GxP/SOX systems per POL-014.

3. Roles and Responsibilities

Role	Responsibility in This Process
DIO (Data Integrity Officer)	Co-owner. Owns the GxP audit trail review programme. Reviews audit trail review records for GxP systems. Escalates ALCOA+ anomalies to Compliance / QA. Approves changes to GxP audit trail configuration (dual approval with CISO).
Information Security Manager	Owns the SIEM monitoring and alert triage programme. Receives and actions gap detection alerts. Co-approves changes to audit trail configuration. Escalates security anomalies detected in audit trail review.
CISO	Dual-approves changes to audit trail configuration on Tier 1 and GxP systems (with DIO). Receives critical security alerts. Notified immediately on audit trail gap detection.
Compliance / QA	Reviews GxP audit trail review records as quality oversight. Responds to ALCOA+ anomaly escalations. Assesses regulatory consequence of

Role	Responsibility in This Process
	audit trail gaps on GxP systems. Signs off GxP audit trail adequacy confirmations.
Data Custodian (IT)	Maintains audit trail technical infrastructure: SIEM log forwarding, storage, retention. Investigates and resolves audit trail gap alerts. Confirms backup integrity of audit trail records.
GxP System Reviewers	Named, trained individuals who conduct routine GxP audit trail reviews per Section 4. Reviewer qualification confirmed by Compliance / QA. Must be different individuals from those whose records they are reviewing (independence requirement).
SOX Control Owners	Conduct or participate in quarterly SOX ITGC audit trail reviews per Section 6.3.

4. GxP Audit Trail Review

GxP audit trail review is a quality activity — it is not a security activity and it is not delegable to IT. The reviewer must be a qualified individual (per Section 4.1) who understands the GxP process being reviewed and can identify meaningful anomalies in the context of that process.

4.1 Reviewer Qualification

An individual may not conduct GxP audit trail reviews for a system until they have completed:

- System-specific GxP training (as confirmed by Compliance / QA and recorded in the VMF per SOP-006)
- GxP audit trail review training — covering ALCOA+ requirements, what constitutes an anomaly for the system, and how to document and escalate findings
- Completion of at least one supervised review alongside a qualified reviewer before conducting independent reviews

Reviewer qualification is recorded in the VMF (Section 15 — Training Records per SOP-006). Compliance / QA confirms qualification before a reviewer's first independent review. Reviewer qualification is reviewed annually.

> **IMPORTANT:** The independence requirement is absolute: a reviewer must not review records that they created, approved, or modified. A system administrator who provisioned access to a system may not review access-related audit trail records for that system. Compliance / QA confirms independence before assigning reviewers to systems.

4.2 Review Scope

Each GxP audit trail review must cover the following record categories — reviewers must confirm that each category has been reviewed and that the applicable ALCOA+ dimensions have been assessed:

Record Category	ALCOA+ Dimensions to Assess	What to Look For
Data creation records	Attributable: is every record attributed to a named, individual user (never generic or shared account)? Contemporaneous: is the timestamp server-generated, accurate to within ±1 second of real time, and in UTC? Original: is the record the first capture of the data, or a	Records attributed to system accounts, shared accounts, or service accounts; timestamp anomalies (clock skew > 1 second, entries out of sequence); entries with no user attribution

Record Category	ALCOA+ Dimensions to Assess	What to Look For
	copy of a copy?	
Data modification records	All of the above, plus: Accurate: are before/after values captured for all modifications? Complete: is the reason for change recorded?	Modifications with no before/after value; modifications with no reason for change; bulk modifications at unusual times; modifications by individuals not authorised for the data type
Data deletion and archival records	Attributable, Contemporaneous, Complete: every deletion must be attributed, timestamped, and the deletion reason recorded.	Deletions without reason; deletions outside normal business processes; records deleted and recreated (indicates alteration attempt); deletion by privileged accounts outside a change record
Electronic approval and signature records	Attributable: is the signature bound to the specific record version being approved? Contemporaneous: does the signature timestamp match the review activity?	Signatures applied in bulk without individual review evidence; signature timestamps inconsistent with documented review timeline; signatures by individuals not listed as qualified reviewers for the step
Access control events	Attributable, Contemporaneous: all access changes attributed to a named individual with accurate timestamp.	Access granted without a corresponding SOP-008 provisioning record; access granted outside business hours without a change record; dormant accounts showing recent access
System configuration changes	Attributable, Contemporaneous, Complete: every configuration change attributed, timestamped, and linked to an approved change record (SOP-007).	Configuration changes without a linked SOP-007 change record; changes made by vendor or service accounts without an associated change ticket; audit trail configuration changes (these must have dual CISO + DIO approval per POL-014)

4.3 Review Cadence

System Type	Review Frequency	Review Period Covered	Reviewer
GxP — active batch processing or clinical data capture	Weekly minimum — more frequent where volume warrants	Prior 7 days	Named GxP Reviewer, independent of records reviewed
GxP — document management and approval workflows	Weekly	Prior 7 days	Named GxP Reviewer

System Type	Review Frequency	Review Period Covered	Reviewer
GxP — system configuration and administration	Weekly	Prior 7 days — focus on configuration events and access changes	DIO or designated qualified reviewer
GxP — archive and read-only systems	Monthly	Prior calendar month	DIO or designated qualified reviewer

4.4 Review Execution and Documentation

Step 01

Pre-Review Preparation

Before conducting a GxP audit trail review: 1. Confirm the audit trail export is complete for the review period — no gaps in the sequence of record IDs 2. Confirm the export is from the authoritative system source — not a cached copy or a secondary report 3. Confirm the reviewer is qualified and independent for this review 4. Note any known events in the period that may generate expected anomalies (e.g., approved emergency changes, system maintenance windows) — these are pre-documented so they are not reported as unexpected findings

Responsible
GxP System Reviewer

Step 02

Review Execution

The reviewer works through each record category (Section 4.2) systematically: For each anomaly identified: — Document: what was found, in which record, at what timestamp, attributed to which user — Classify: Unexplained (no known legitimate reason) / Explained (consistent with a known approved activity, but not pre-documented) / Expected (pre-documented in Step 1) — Unexplained anomalies: escalated to DIO and Compliance / QA immediately — do not defer to end of review — Explained anomalies: documented in the review record with the explanation and the evidence supporting the explanation The review is complete when all record categories have been covered for the full review period. Partial reviews are not acceptable — if the review period cannot be fully covered in a single session, the review is continued in the next session with the continuity explicitly noted.

Responsible
GxP System Reviewer

Step 03

Review Record Completion

The reviewer completes the GxP Audit Trail Review Record (a GxP quality record): Mandatory fields: — System name and ID — Review period (start and end timestamp) — Reviewer name (printed) and qualification reference — Date of review — Record categories reviewed (checklist) — Total number of records reviewed — Number of anomalies identified: none / [N] — with classification for each — For each anomaly: description, classification, evidence supporting explanation or escalation reference — Reviewer declaration: 'I confirm this audit trail review is complete for the stated period, all anomalies have been documented, and unexplained anomalies have been escalated.' — Reviewer signature and date — DIO or Compliance / QA countersignature (for weekly GxP

Responsible
GxP System Reviewer / DIO

reviews: DIO countersigns monthly batch; for any review with unexplained anomalies: Compliance / QA countersigns immediately) Review records are GxP quality records: version-controlled, filed in the VMF (Section 14 — Audit Trail Review Records per SOP-006), and retained for the life of the system plus the applicable GxP retention period per POL-002.

4.5 Unexplained Anomaly Response

An unexplained anomaly in a GxP audit trail is a potential data integrity event. The response is immediate — it does not wait for the end of the review cycle.

Anomaly Type	Immediate Action	DIRA Assessment	SOP-012 Trigger?
Modification without reason for change or before/after values	DIO and Compliance/QA notified within 1 hour. Affected record quarantined (no further use in GxP decisions until assessed).	DIRA assessment for ALCOA+ Accurate and Complete. Assess whether affected record can be reconstructed from other evidence.	Yes — if modification appears unauthorised or malicious.
Record attributed to shared or service account	DIO and Compliance/QA notified within 1 hour. Data Custodian IT confirms account type and investigates how access was obtained.	DIRA assessment for ALCOA+ Attributable. The record cannot be attributed to an individual and may not be acceptable as GxP evidence.	Yes — if account was not authorised for GxP access or if access was outside a change record.
Timestamp anomaly (clock skew > 1 second, out-of-sequence entries)	DIO and Data Custodian IT notified within 1 hour. NTP synchronisation confirmed. Extent of affected records assessed.	DIRA assessment for ALCOA+ Contemporaneous. Clock skew affects the integrity of timestamps across the review period.	Only if skew is persistent and cannot be explained by an NTP fault — persistent clock manipulation is a security indicator.
Configuration change without linked SOP-007 change record	DIO and Information Security Manager notified immediately.	DIRA assessment for ALCOA+ Complete. Unauthorised configuration change may affect the validated	Yes — unauthorised configuration change is

Anomaly Type	Immediate Action	DIRA Assessment	SOP-012 Trigger?
	Change assessed against SOP-007 change register for the period.	state.	a security event. Also triggers SOP-006 validated state deviation assessment .
Audit trail gap — missing entries for a time period	Immediate — see Section 5.3 (Audit Trail Gap Detection and Response). This is the highest-priority anomaly category.	DIRA assessment for ALCOA+ Complete. A gap means records for the period cannot be verified as complete.	Yes — always. Audit trail gap is treated as a simultaneous SOP-003 and SOP-012 event.

5. SIEM Monitoring and Alert Triage

The SIEM receives audit trail records from all Tier 1/2 and GxP/SOX systems within 5 minutes of generation (per POL-014). The monitoring programme applies detection rules to those records and generates alerts that must be triaged by Information Security.

5.1 Alert Categories and Priority

Category	Alert Examples	Priority	Triage SLA	Escalation
Category 1 Audit Trail Integrity	Log forwarding failure to SIEM (>15 minutes gap in log stream); audit trail gap detected in GxP or Tier 1 system; hash chain validation failure on systems with cryptographic integrity controls; audit trail configuration change outside approved change window.	Critical	15 minutes from alert to initial triage. Information Security Manager paged immediately .	DIO and CISO notified within 15 minutes. Simultaneous SOP-003 (DIRA) and SOP-012 (security incident) assessment. GxP system quarantined pending investigation.
Category 2 Privileged	Privileged account access outside an approved	High	30 minutes from alert to	CISO and Information Security Manager notified. SOP-009

Category	Alert Examples	Priority	Triage SLA	Escalation
Access Anomaly	change or maintenance window; service account interactive login; PAM session credentials used outside the PAM workflow; GxP privileged access without linked change record.		initial triage.	procedure reviewed. If GxP system: DIO and Compliance/QA notified. SOP-012 assessment if unauthorised access confirmed.
Category 3 Authentication Anomaly	Impossible travel (authentication from two geographically incompatible locations within a short time window); ≥5 failed authentication attempts within 10 minutes on Tier 1/GxP system; authentication from a new country not in the user's travel history; MFA bypass attempt.	High	30 minutes from alert to initial triage.	Information Security Manager notified. Account suspension if impossible travel confirmed. SOP-012 assessment.
Category 4 Data Exfiltration Indicator	Abnormal outbound data volume from Tier 1/GxP system; mass download or export of records outside normal business patterns; data access by account with an active leaver flag; transfer of GxP records to unapproved external storage.	High	30 minutes from alert to initial triage.	Information Security Manager and DPO notified (personal data potential breach). CISO for P1 threshold volumes. SOP-012 assessment.
Category 5 Vulnerability and Exploitation	Known exploit attempt signature detected; attack pattern consistent with CVE against an in-scope system; web application attack (SQLi, XSS) attempt on Tier 1 system.	Medium	2 hours from alert to initial triage.	Information Security Manager notified. IS Risk Register entry if confirmed vulnerability. SOP-012 if exploitation confirmed.

5.2 Alert Triage Process

Step 01	**Alert Receipt and Initial Assessment** On receipt of a SIEM alert, the on-call Information Security analyst: 1. Confirms the alert is not a known false positive (false positive exclusion list maintained by Information Security Manager and reviewed quarterly) 2. Assesses the alert category and priority per Section 5.1 3. For Category 1 alerts: escalates to Information Security Manager immediately — does not attempt to resolve independently 4. For Category 2/3/4 alerts:	Responsible **On-call Information Security Analyst**

	begins initial investigation to confirm or deny the alert indicator 5. Opens a triage record in the ITSM platform with the alert timestamp, category, and initial assessment The triage SLA clock starts at the alert timestamp — not when the analyst noticed it. Analysts reviewing alerts must process the queue in priority order.	
Step 02	**Triage Investigation** Initial investigation to determine: is this a true positive or a false positive? Investigation approach by category: Category 1 (Audit Trail Integrity): confirm gap in SIEM log stream. Query the source system directly — is the system generating logs? Is the forwarding agent running? If the source system is not generating logs, this is an immediate SOP-003/SOP-012 event. Category 2 (Privileged Access): confirm in PAM platform whether a session was active at the time of the alert. Review the SOP-007 change register for an associated change record. If no change record and no PAM session: unauthorised privileged access confirmed. Category 3 (Authentication): confirm the authentication events in the identity provider. Cross-reference with travel or location records if available. If impossible travel confirmed: account compromise presumed until disproved. Category 4 (Data Exfiltration): confirm data volume against baseline for the user/system. Identify destination. If destination is outside approved storage/systems: escalate to Information Security Manager immediately. Category 5 (Vulnerability): confirm whether the exploit attempt was blocked by existing controls. Identify affected system and check vulnerability status.	Responsible **Information Security Analyst / Information Security Manager**
Step 03	**Triage Disposition** On completion of initial investigation, the analyst records a disposition: True Positive — Escalate: the alert represents a real event requiring further action per the escalation column in Section 5.1. ITSM ticket escalated per the category. SOP-012 and/or SOP-003 triggered as applicable. True Positive — Managed: the alert is real but within expected parameters and being addressed within normal processes (e.g., a Category 5 alert for an attempted exploit that was blocked by the WAF and the system is already patched). False Positive — Known: the alert matches an entry in the false positive exclusion list. ITSM ticket closed with false positive reference. False Positive — New: the alert is assessed as a false positive not previously identified. ITSM ticket closed with false positive rationale. False positive exclusion list updated by Information Security Manager after review. All dispositions are recorded in the ITSM platform. Closed tickets are not deleted — they are retained per POL-002.	Responsible **Information Security Analyst**

5.3 Audit Trail Gap Detection and Emergency Response

IMPORTANT: An audit trail gap — a period where records should exist but do not — is the most serious audit trail event. For GxP systems, a gap means records for the affected period cannot be verified as complete, accurate, or authentic. This is a simultaneous data integrity event (SOP-003) and potential security event (SOP-012). The response is immediate.

Step 01	**Gap Detection** Audit trail gaps are detected through two mechanisms: Automated — SIEM log stream monitoring: the SIEM monitors the sequence and timing of log records from each source system. A gap in the expected log stream (>15 minutes without records from a system that should be generating them) generates an immediate Category 1 alert per Section 5.1. Manual — Audit trail review: during the GxP audit trail review (Section 4), the reviewer confirms that record IDs are sequential and complete for the review period. A missing ID sequence or a time period with no records is a gap finding. In either case, the gap is treated as a Critical event from the moment it is identified.	Responsible **SIEM / GxP Reviewer**
Step 02	**Immediate Notification — Within 15 Minutes** On identification of an audit trail gap on any GxP or Tier 1 system: 1. Information Security Manager paged immediately 2. DIO notified within 15 minutes 3. CISO notified within 15 minutes 4. For GxP systems: Compliance / QA Director notified within 15 minutes The system involved is placed under enhanced monitoring immediately — all subsequent records reviewed in real time until the gap is resolved. For GxP systems: Compliance / QA Director assesses whether GxP activities on the affected system should be suspended pending investigation. If there is any doubt: suspend. The risk of continuing GxP activities on a system with a compromised audit trail exceeds the operational cost of suspension.	Responsible **Information Security Manager / DIO / CISO / Compliance QA**
Step 03	**Gap Investigation** Data Custodian (IT) and Information Security investigate the gap simultaneously: Technical investigation: — Is the source system running and generating logs? If not: SOP-015 operational incident process activated in parallel. — Is the SIEM forwarding agent running on the source system? If not: is this a technical failure or a deliberate disablement? — Are the records missing from the source system's local audit trail, or only from the SIEM? If only from SIEM: log forwarding failure (technical). If missing from source system: data integrity event. — What was the last record before the gap, and what is the first record after? Is there evidence of system modification during the gap window? Security investigation (parallel): — Were there any privileged access events in the period before the gap? — Were there any configuration changes to the audit trail or logging components? — Does the gap correlate with any other anomalies in the SIEM for this system? Investigation timeline: initial findings reported to DIO, CISO, and Compliance / QA within 2 hours of gap detection.	Responsible **Data Custodian (IT) / Information Security**

Step 04	**Gap Classification and Response** Based on investigation, the gap is classified: Classification A — Technical failure (SIEM forwarding failure only): records exist in source system's local audit trail and are complete. SIEM forwarding agent failed or was temporarily disrupted by a known technical cause. Records backfilled from source system to SIEM. Response: Data Custodian IT restores forwarding and backfills records. DIO confirms source system audit trail is complete and intact. Review record updated: gap explained, evidence documented. SOP-003 DIRA not required. SOP-012 not required unless the forwarding failure was caused by a suspicious event. Classification B — Source system gap (records missing from source): records are absent from the source system's audit trail for the gap period. Response: this is always a potential ALCOA+ Completeness failure and a potential security event. Both SOP-003 DIRA (immediate assessment) and SOP-012 (security incident, Category 1 escalation) are triggered simultaneously. GxP activities on the affected system suspended. Compliance / QA Director leads regulatory consequence assessment. DIO leads data integrity assessment. CISO leads security investigation. Classification and response rationale documented in the ITSM incident record, DIRA assessment record (SOP-003), and VMF (GxP systems).	Responsible **DIO / CISO / Compliance QA**

6. Periodic Review Programme

The periodic review programme provides structured oversight of audit trail coverage, SIEM alert handling effectiveness, and control adequacy across all system tiers. Reviews are documented and retained as controlled records.

Review Type	System Scope	Frequency	Review Content	Owner	Evidence Required
Continuous + Monthly Report	Tier 1 and all GxP systems	Continuous monitoring (automated); Monthly written report	SIEM alert summary: alerts by category, triage outcomes, false positive rate. GxP audit trail review summary: reviews completed, anomalies by classification, escalations. Any open gap investigations. Coverage confirmation: all required systems are forwarding to SIEM.	Information Security Manager + DIO	Monthly monitoring report. Sign by Informati Security Manager an DIO. Filed ir ITSM. GxP review recor filed in VMF
Quarterly SOX ITGC Review	All SOX ITGC systems	Quarterly	Access review results confirmed (per SOP-008). Privileged access review results confirmed (per SOP-009). Audit trail coverage confirmed: all	Information Security Manager + Finance Controller	Quarterly S ITGC audit t review recor Signed by Information Security

Review Type	System Scope	Frequenc y	Review Content	Owner	Evidence Required
			SOX ITGC systems forwarding to SIEM. SIEM retention for SOX ITGC logs confirmed at 7 years minimum. Any open findings or exceptions documented.		Manager an Finance Controller. Retained 7 y per POL-00
Monthly Tier 2 Review	Tier 2 non-GxP, non-SOX systems	Monthly	SIEM alert summary for Tier 2 systems. Access review status confirmation. Dormant account check. Any anomalies identified.	Information Security Manager	Monthly revi record filed ITSM.
Quarterly Tier 3/4 Review	Tier 3 and Tier 4 systems	Quarterly	Confirm log forwarding is configured and operational (where applicable). Review any SIEM alerts for these systems. Annual coverage review: confirm Tier 3/4 systems not forwarding to SIEM have local audit trail retention in place.	Information Security	Quarterly re record filed ITSM.
Annual Log Coverage Review	All systems	Annual	Confirm all Tier 1/2/GxP/SOX systems are forwarding to SIEM. Confirm SIEM retention periods meet POL-014 requirements. Identify any Tier 1/2/GxP/SOX systems added since the last review that are not yet forwarding. Identify systems decommissioned since last review — confirm log archive integrity.	Information Security Manager + DIO	Annual Log Coverage Review Rep Signed by Information Security Manager an CISO. Findi raised as IS Register ent (SOP-010).

7. Immutability Verification

POL-014 requires that audit trail records are technically immutable — unmodifiable and undeletable by any user including system administrators. For systems using cryptographic hash chaining as the immutability mechanism, periodic verification of the hash chain integrity provides positive assurance that records have not been tampered with since they were written.

System Type	Immutability Mechanism	Verification Method	Verification Frequency	Owner
Systems with cryptographic hash chaining	Each audit trail record contains a hash of its own content plus the hash of the preceding record — creating a tamper-evident chain.	Hash chain validation: a script or tool verifies that every record in the chain produces the expected hash when re-hashed, and that each record's predecessor hash matches the actual hash of the preceding record. Any break in the chain indicates tampering or corruption.	Monthly for GxP and Tier 1 systems. Quarterly for Tier 2 systems.	Data Custodian (IT) executes. DIO reviews results for GxP systems.
Systems with WORM storage or immutable cloud audit services	Write-Once Read-Many storage — records cannot be overwritten or deleted at the storage layer.	Storage integrity confirmation: confirm WORM locks are applied and that no delete or overwrite operations have succeeded. Cloud audit service: confirm immutability status in service configuration console.	Quarterly for all systems using this mechanism.	Data Custodian (IT).
Systems with append-only database tables	Database schema configured so audit trail tables have no UPDATE or DELETE permissions for any database role including DBA.	Permission audit: confirm that no role has UPDATE or DELETE permissions on audit trail tables. Test: attempt UPDATE as DBA — should fail. Confirm test result is documented.	Quarterly permission audit. Annual DBA test.	Data Custodian (IT). DIO countersigns for GxP systems.
SIEM storage	SIEM storage is itself immutable — records forwarded to SIEM are not modifiable.	SIEM vendor attestation of immutability, supplemented by: confirm no records have been deleted from the SIEM audit log for the period under review. Log volume reconciliation: expected vs. received record counts.	Monthly volume reconciliation. Annual attestation review.	Information Security Manager.

Immutability verification results are documented and retained. For GxP systems: verification records are filed in the VMF (Section 14) and are quality records. A failed immutability verification — a broken hash chain, a successful DELETE on an append-only table, or unexplained record deletion from SIEM — is a Critical audit trail integrity event and triggers the gap detection response process (Section 5.3, Classification B).

8. Related Documents

Document ID	Title	Relationship
POL-014	Audit Trail and Monitoring Policy	Parent policy — this SOP operationalises POL-014
POL-003	Data Integrity Policy	ALCOA+ requirements are the interpretive framework for all GxP audit trail review; audit trail integrity is a data integrity control
POL-005	System Classification Policy	System tier determines review cadence, SIEM forwarding requirement, and immutability verification frequency
POL-006	CSV Policy	GxP audit trail review records are quality records filed in the VMF per SOP-006; audit trail configuration validated in IQ; immutability and cannot-be-disabled requirements tested in OQ
POL-007	Change Management Policy	Changes to audit trail configuration require dual CISO + DIO approval and are Normal changes minimum under SOP-007
POL-009	PAM Policy	Privileged access anomalies detected in audit trail review are escalated via SIEM Category 2 alerts; PAM session recordings are forensic corroboration for privileged access audit trail entries
POL-010	IS Risk Policy	Audit trail gaps, immutability failures, and persistent monitoring anomalies are IS risks entered in the IS Risk Register via SOP-010
POL-012	Incident Response Policy	Audit trail integrity events (Classification B gaps, immutability failures, unauthorised configuration changes) are simultaneous SOP-003 and SOP-012 events
POL-015	IT Service Management Policy	Log forwarding failures investigated under SOP-015 operational incident process in parallel with audit trail gap response
SOP-003	Data Integrity Process	DIRA is triggered for all Classification B audit trail gaps and ALCOA+ anomalies; this SOP is the primary feeder of DIRA trigger events for GxP systems
SOP-006	Computer System Validation Process	Audit trail review records are VMF quality records; audit trail configuration validated in IQ/OQ; immutability verified in OQ
SOP-007	Change Management Process	Audit trail configuration changes are Normal changes with dual approval
SOP-009	PAM Process	Category 2 privileged access anomalies from

Document ID	Title	Relationship
		SIEM feed into SOP-009 response; PAM session recordings confirm or contradict audit trail entries
SOP-010	IS Risk Assessment Process	Annual log coverage review generates IS Risk Register entries; monitoring programme effectiveness is an IS risk assessment input
SOP-012	Security Incident Response Process	Audit trail integrity events simultaneously activate SOP-003 and SOP-012; CISO notified within 15 minutes of Category 1 alert
SOP-015	ITSM Process	Log forwarding infrastructure failures are operational incidents under SOP-015; parallel processes where the cause is unclear
EXC-001	Exception Management Policy and Register	Exceptions to audit trail review cadence (e.g., delayed review due to personnel availability) are Category 4 exceptions under EXC-001; maximum 3 months; disabling audit trail review on GxP systems is non-approvable
RACI-001	Roles, Responsibilities and Authorities	Accountability for audit trail and monitoring governance

— End of Document —

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